

SW1
TS-1088-AR02016

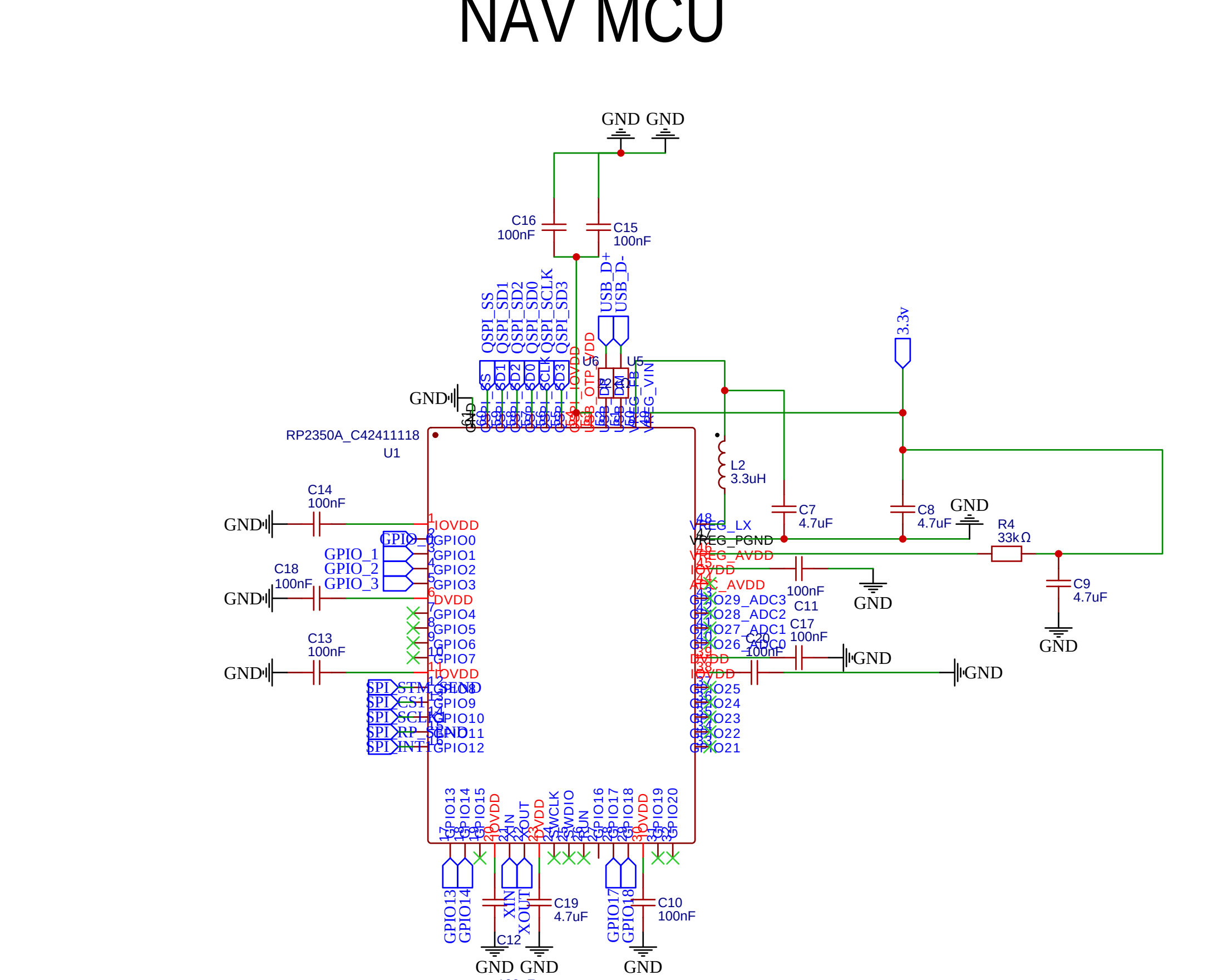
QSPI_SS

2

1

GND

NAV MCU



The schematic diagram illustrates the electrical connections for the NAV MCU. The central component is the RP2350A_C42411118 (U1), which is connected to various peripheral components. The power supply section includes a 3.3V regulator (L2) and a 4.7uF capacitor (C7) connected to the VDD and GND pins. The USB interface is connected to the USB_D+ and USB_D- pins, with a 100nF capacitor (C15) connected to GND. The SPI interface is connected to the SPI_SS, SPI_MOSI, SPI_MISO, and SPI_SCLK pins, with a 100nF capacitor (C16) connected to GND. The I2C interface is connected to the I2C_SDA and I2C_SCL pins, with a 100nF capacitor (C17) connected to GND. The UART interface is connected to the UART_TX and UART_RX pins, with a 100nF capacitor (C18) connected to GND. The GPIO pins are connected to various external components, including a 3.3V regulator (L2) and a 4.7uF capacitor (C7) connected to the VDD and GND pins. The schematic also shows the connection of the RP2350A_C42411118 to the USB interface, SPI interface, I2C interface, and UART interface. The power supply section includes a 3.3V regulator (L2) and a 4.7uF capacitor (C7) connected to the VDD and GND pins. The schematic also shows the connection of the RP2350A_C42411118 to the USB interface, SPI interface, I2C interface, and UART interface.

GPS

Crystal

The diagram illustrates a crystal oscillator circuit. It features a central crystal component labeled 'U8 12MHz'. The crystal has four pins: pin 1 is connected to GND, pin 2 is connected to GND, pin 3 is connected to a network of components, and pin 4 is connected to a network of components. The network connected to pin 3 consists of a capacitor 'C23 15pF' in series with a resistor 'R6 1kΩ', which is then connected to the output 'XOUT'. The network connected to pin 4 consists of a capacitor 'C22 15pF' in series with the input 'XIN'. Both input and output are represented by blue trapezoidal symbols. Ground connections are shown as three horizontal lines of decreasing width.

MAGNOMETER

The diagram shows a magnetometer module (U16: IST8310 C2683055) connected to a 3.3V supply and GND. The module has pins for SCL, AVDD, GND, NC, I2C, CAD0, CAD1, VPP, and NC. It is connected to a 3.3V supply via a 33kΩ resistor (R7) and to GND via a 4.7uF capacitor (C38).

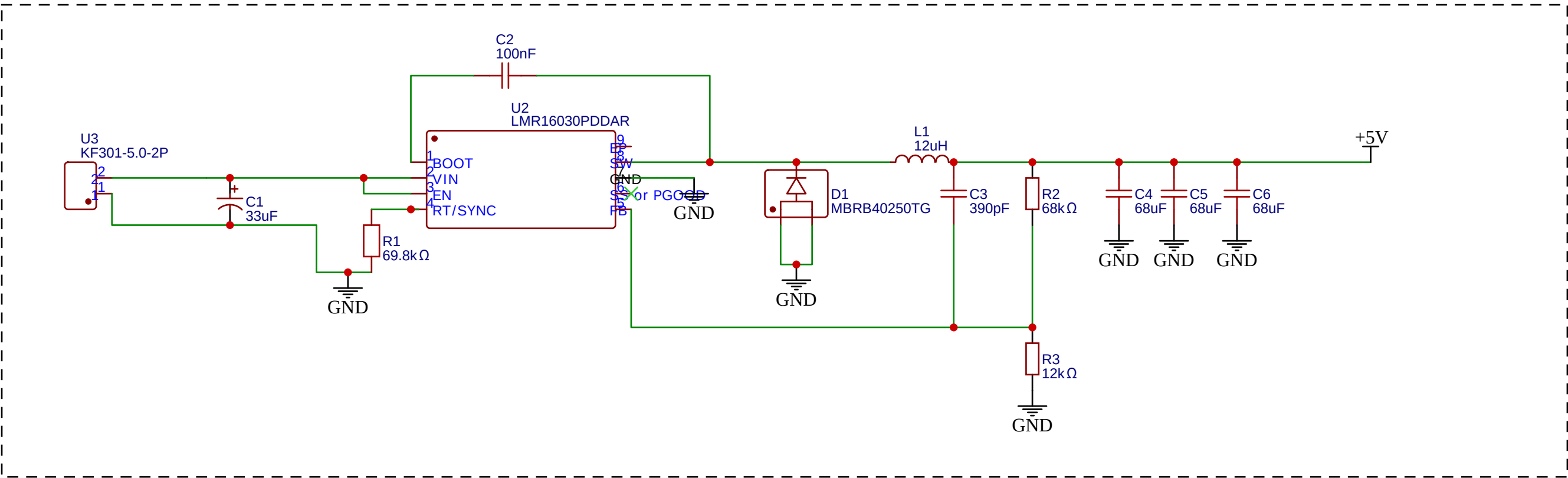
Barometer

The diagram shows the wiring for a BMP280 barometer module (labeled U17). The module is represented by a red rectangle with pins numbered 1 through 8. The connections are as follows:

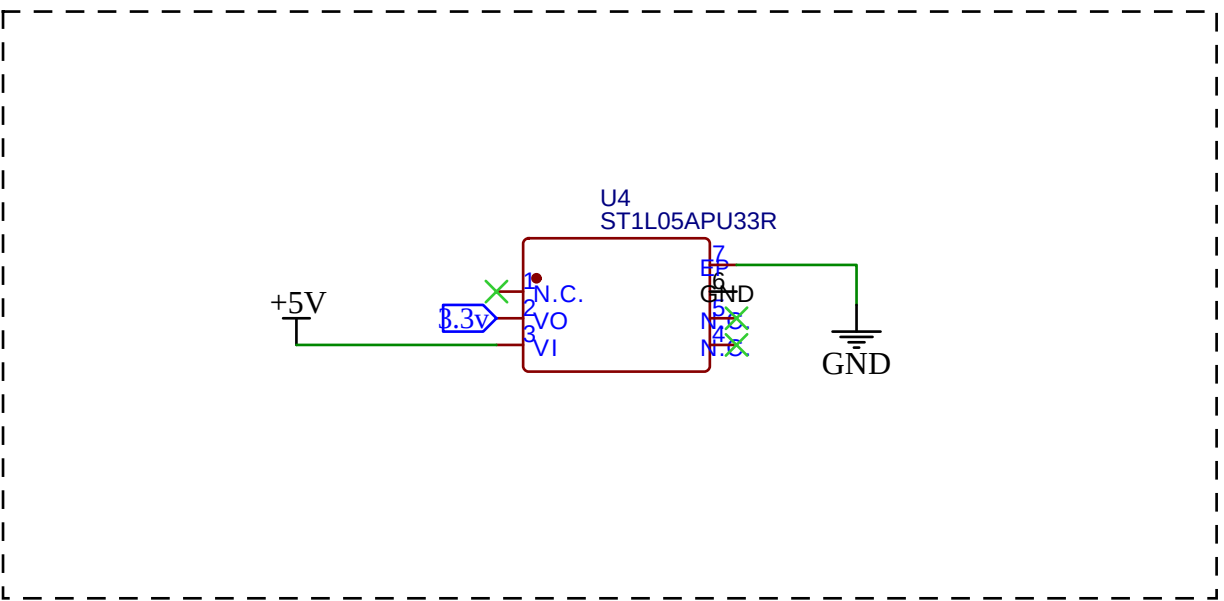
- Pin 1 (GND) is connected to a common ground.
- Pin 2 (VCC) is connected to a 3.3V supply.
- Pin 3 (GND) is connected to a common ground.
- Pin 4 (VCC) is connected to a 3.3V supply.
- Pin 5 (GND) is connected to a common ground.
- Pin 6 (VDDIO) is connected to a 3.3V supply.
- Pin 7 (GND) is connected to a common ground.
- Pin 8 (SDO) is connected to a common ground.

The module is labeled "U17 BMP280".

7-40v to 5v



5 to 3.3v



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